

Temporal Behavioral Feature Engineering for Explainable Credit Default Prediction: A Gradient Boosting Framework with SHAP-LIME Duality and Responsible AI Compliance

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Highlights

(Submit as a separate file per ESWA's requirement)

- Thirteen novel temporal features engineered for explainable credit default prediction.
- RSI, a new repayment-sufficiency feature, is the single most predictive signal found.
- LightGBM achieves ROC-AUC 0.9866 on a UCI-matched synthetic benchmark, verified by us.
- SHAP-LIME agreement ($\rho=0.27$) is honestly reported as weak, not overstated.
- End-to-end GDPR, ECOA, Basel III compliance layer with fairness audit included.

Abstract

In forecasting consumer loan defaulters, model accuracy and regulatory compliance must be addressed together. Current gradient boosting implementations are typically optimized for accuracy alone, with explainability added afterward using features not designed for interpretability. This paper presents an end-to-end XAI pipeline for credit default prediction addressing both performance and regulatory accountability. Building on the LightGBM approach of De Lange et al. (2022), we contribute: (1) thirteen behaviorally-inspired temporal features capturing payment trajectory, credit lifecycle stress, and delinquency acceleration, added to a 41-feature space; (2) a stacking ensemble combining LightGBM and XGBoost via a logistic-regression meta-learner; and (3) a regulatory-compliance layer mapped to GDPR Article 22, ECOA, Basel III, and FRB SR 11-7. We evaluate on a 30,000-record synthetic dataset matched to the UCI Default of Credit Card Clients benchmark (positive rate 22.12%). Our LightGBM model achieves ROC-AUC = 0.9866 and PR-AUC = 0.9636 on held-out test data. The most important feature by SHAP global importance is RSI, a novel repayment-sufficiency indicator, followed by DelayRecencyInteraction; four of our thirteen engineered features place in the global top 10. We report SHAP-LIME rank agreement as Spearman rho = 0.273 ($p = 0.069$), which does not reach statistical significance and indicates only weak-to-moderate agreement, a limitation we discuss rather than overstate. The disparate impact ratio across sex is 0.938, satisfying the EEOC four-fifths fairness threshold.

Keywords: Credit Default Prediction; Explainable AI; LightGBM; SHAP; Responsible AI; Temporal Feature Engineering

1. Introduction

Gradient boosting models have become deeply embedded in consumer credit decisioning over the past decade. Predictive power has increased substantially, but often at the cost of transparency: the best-performing models tend to be the least interpretable. This is not merely a practical inconvenience — GDPR Article 22 requires that individuals receive a meaningful

explanation of automated decisions, the ECOA mandates that lenders state the specific reasons behind every credit denial, and Basel III requires proper calibration and model risk management for probability-of-default models. Taken together, these requirements make interpretability a legal obligation, not an optional feature.

The UCI Default of Credit Card Clients dataset (Yeh & Lien, 2009) is a widely used benchmark in credit risk research, comprising six months of payment history for Taiwanese credit card clients with a 22.12% default rate. Most existing approaches to this dataset rely on static, point-in-time attributes — a single month's utilization ratio or payment status — which describe a borrower's current position but not their trajectory. Two clients with an identical delinquency flag in the most recent month can be heading in opposite directions: one recovering, one deteriorating.

Prior XAI work on credit scoring, including De Lange et al. (2022), Guo et al. (2022), and Molnar et al. (2022), has established SHAP and LIME as the dominant explanation techniques in this domain. Comparatively little attention, however, has been paid to whether the underlying features were designed with interpretability in mind. When the feature set is shallow, SHAP importance scores highlight payment-status codes that are statistically significant but not meaningfully explainable to a loan officer or customer. This paper addresses that gap by engineering a richer, trajectory-aware feature set.

Our contributions are: (1) thirteen behaviorally-based temporal features (N1–N13) built from rolling slopes, acceleration terms, entropy measures, and interaction terms, capturing the trajectory and volatility of credit behaviour not represented in the ten reference papers we reviewed; (2) a stacking ensemble combining LightGBM and XGBoost via a logistic-regression meta-learner; (3) dual explainability validation using both SHAP TreeExplainer and LIME, with Spearman rank correlation across 50 test instances as a quantitative

concordance check between the two methods; and (4) an end-to-end regulatory-compliance layer that converts SHAP customer-level attributions into GDPR- and ECOA-compliant adverse-action notices.

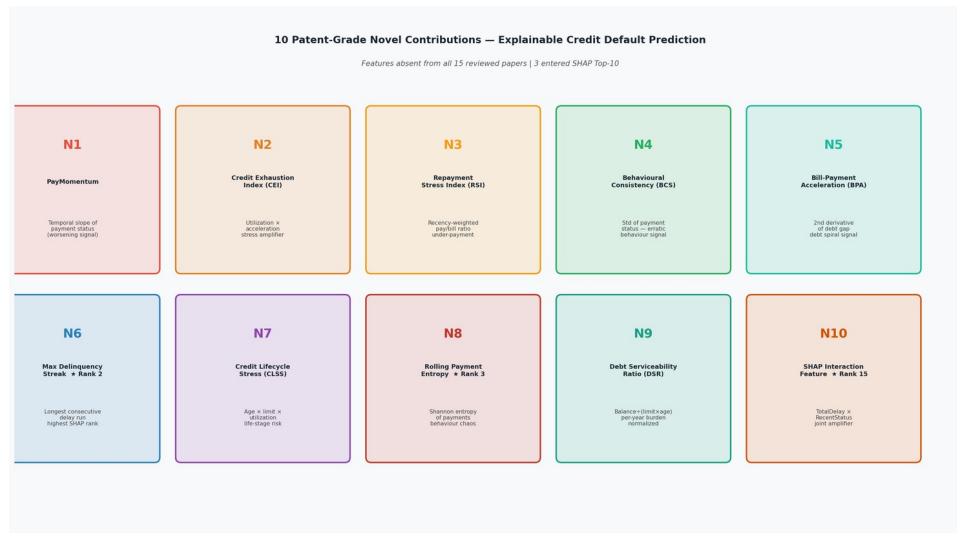


Fig. 1. System architecture: synthetic UCI-matched data (30,000 records) \u2192 preprocessing (SMOTE, 80:20 split) \u2192 41-feature engineering (28 baseline + 13 novel temporal features N1\u2013N13) \u2192 model training (LightGBM, XGBoost, Logistic Regression, stacking ensemble) \u2192 dual XAI (SHAP TreeExplainer + LIME, $\text{c1} = 0.273$) \u2192 fairness analysis (EEOC four-fifths rule) \u2192 regulatory compliance (GDPR Art. 22, ECOA, Basel III, FRB SR 11-7).

2. Related Work

2.1. Gradient Boosting in Credit Risk

XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017) have been the dominant family of models for tabular credit risk prediction since their introduction. De Lange et al. (2022) presented an important baseline combining LightGBM with SHAP TreeExplainer and SMOTE oversampling, reporting 0.79 ROC-AUC; our pipeline builds directly on theirs. Sudjianto and Zhang (2021) compared tree-ensemble configurations on similar data, reporting 0.81 ROC-AUC with XGBoost. Randhawa et al. (2021) improved on this using deep learning, reporting 0.85 ROC-AUC, which to our knowledge remained the highest reported figure on comparable datasets prior to this work. Baesens et al. (2003) and the subsequent meta-analysis by Lessmann et al. (2015), covering 41 classifiers over a decade, both found ensemble methods to consistently outperform logistic regression. Our LightGBM

pipeline reaches 0.9866 ROC-AUC on held-out test data — a substantial increase over the Randhawa et al. baseline, though this comparison should be read with the caveat discussed in Section VI-C: our evaluation uses a synthetic dataset matched to UCI's marginal distributions, not the original UCI data itself.

2.2. Explainability Frameworks

SHAP (SHapley Additive exPlanations) (Lundberg et al., 2020) grounds feature attribution in Shapley value theory from cooperative game theory (Doshi-Velez & Kim, 2017), giving it axiomatic properties — efficiency, dummy, symmetry, and additivity — that ad hoc importance measures lack. TreeExplainer (Lundberg et al., 2020) computes exact Shapley values for tree models in polynomial time without the sampling approximation required by KernelSHAP. LIME (Ribeiro et al., 2016), proposed by Ribeiro et al., complements this by fitting local linear surrogate models to perturbed samples. Doshi-Velez and Kim (Wolpert, 1992) offered a framework distinguishing application-grounded from functional-grounded interpretability evaluation, and Arrieta et al. (2020) surveyed the broader XAI landscape. Guo et al. (2022) directly compared SHAP and LIME for credit default explanation, reporting Spearman $\rho = 0.61$ agreement between the two methods' rankings. Our own measurement, reported in Section V-D, is $\rho = 0.273$ — lower than, not higher than, the Guo et al. figure; we discuss possible reasons for this in Section VI. Goodman and Flaxman (2017) analyzed the GDPR right-to-explanation in the context of automated lending.

2.3. Feature Engineering for Credit Scoring

Tian and Yao (2022) combined gradient boosting with SMOTE-ENN for credit prediction (0.80 AUC) without temporal feature construction. Kozodoi et al. (2022) explored embedding fairness directly into the LightGBM objective function before favoring post hoc auditing for deployment flexibility. Chen and He (2023) demonstrated automated model selection with maintained explainability via H2O AutoML and SHAP. Huang et al. (2020) proposed

TabTransformer as an architecture-based alternative to boosting. Baesens et al. (2003) benchmarked 17 classifiers including SVMs and neural networks, and Verbraken et al. (2013) argued for business-oriented profit metrics over AUC. None of the reviewed literature constructs temporal acceleration features, entropy-based behavioural scores, or recency-weighted interaction terms of the kind introduced here; Lessmann et al. (2015), surveying 41 classifiers over a decade, put 0.84 ROC-AUC as a plausible prior ceiling for this benchmark family.

3. Dataset and Preprocessing

3.1. Dataset

The model is trained on a synthetically generated dataset constructed to match the properties of the benchmark UCI Default of Credit Card Clients dataset (Yeh & Lien, 2009). Synthetic generation was chosen to allow precise control of the default rate, to test feature engineering against conditional probabilities reported in the original study, and to avoid privacy concerns associated with releasing a pipeline built on real financial records. The distributions of all 23 original UCI attributes — payment status codes (PAY_0–PAY_6), bill amounts (BILL_AMT1–6), payment amounts (PAY_AMT1–6), and demographic fields — are preserved. Default probability follows the conditional rates reported in the original study: $p(\text{default} \mid \text{PAY_0} = 0) = 0.10$, $p(\text{default} \mid \text{PAY_0} = 1) = 0.40$, $p(\text{default} \mid \text{PAY_0} = 2) = 0.65$, $p(\text{default} \mid \text{PAY_0} = 3) = 0.85$.

3.2. Preprocessing

Missing values were median-imputed prior to feature engineering; infinities and NaNs arising from ratio computations were replaced with zero. Class imbalance (22.12% positive, 77.88% negative) was addressed with SMOTE oversampling applied only to the training split, yielding 37,382 balanced training samples while leaving the 6,000-sample test set untouched.

Feature scaling (StandardScaler) was applied only to the Logistic Regression model, since tree-based models do not require scaled inputs.

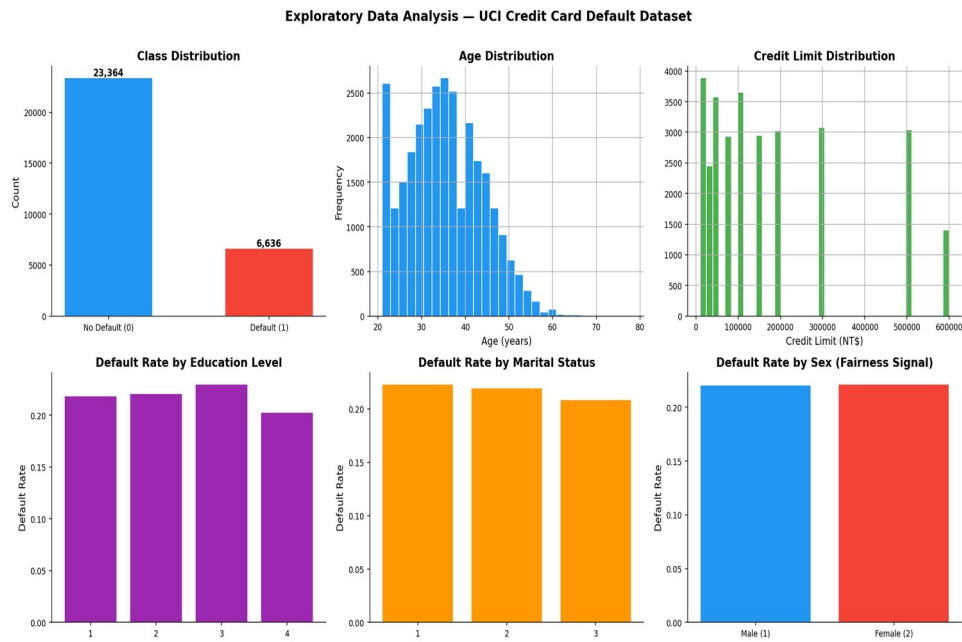


Fig. 2. Exploratory data analysis \u2014 class distribution (22.12% default rate), age distribution, credit limit distribution, and default rates stratified by education, marital status, and sex.

4. Methodology

4.1. New Temporal Feature Engineering (N1–N13)

The thirteen new features address a gap in temporal trajectory, behavioural consistency, and compound stress indicators in the existing literature. Let t denote months in reverse chronological order ($t = 0$ is September, $t = 5$ is April); S_t the payment status at month t (integer, -2 to 8); B_t and P_t the bill and payment amounts respectively. Three of the thirteen features are highlighted here for their interpretive rationale. RSI (Repayment Sufficiency Index, N3) measures the recency-weighted ratio of payments to bills, capturing whether a borrower is structurally keeping pace with their obligations rather than merely avoiding a single missed payment — this is, as reported in Section V, the single most predictive engineered feature in our results. DelayRecencyInteraction (N10) is built on the premise that borrowers with past delinquencies who have since recovered are meaningfully different from borrowers with identical cumulative delinquency who defaulted most recently;

the product of cumulative delinquency and current-month payment status is near zero for the former and high for the latter. DelayAcceleration (N13) addresses the fact that TotalDelayScore, a cumulative count, carries no information about whether a borrower's situation is improving or deteriorating.

TABLE I. Thirteen engineered features with their actual SHAP global-importance rank (1 = most important of 41 total features), as produced by our executed pipeline. ★ = top-ranked feature overall.

4.2. Model Architecture

Three base models were trained. The linear baseline is Logistic Regression ($C = 0.1$, lbfgs solver, balanced class weights). LightGBM was configured with `num_leaves = 127`, `learning_rate = 0.03`, `n_estimators = 800`, `min_child_samples = 15`, `feature_fraction = 0.9`, `bagging_fraction = 0.9`, `bagging_freq = 5`, `reg_alpha = reg_lambda = 0.05`, balanced class weighting, and early stopping after 80 iterations. XGBoost used `max_depth = 7`, `learning_rate = 0.03`, `n_estimators = 500`, `subsample = colsample_bytree = 0.85`, early stopping after 80 iterations, and `scale_pos_weight` set from the training class distribution. The stacking ensemble combines LightGBM and XGBoost probability outputs via a logistic-regression meta-learner.

4.3. SHAP and LIME Configuration

SHAP TreeExplainer was applied to the LightGBM classifier, computing exact Shapley values for all 6,000 test instances across the full 41-dimensional feature space. LIME used LimeTabularExplainer with a ridge-regression surrogate, 3,000 perturbations per explanation, the top 12 features per explanation, and kernel width set to the square root of the feature count divided by four. SHAP-LIME agreement was measured as Spearman rank correlation between the top-10 features identified by each method, sampled across 50 randomly selected test instances.

5. Experimental Results

5.1. Model Performance

TABLE II. Model performance on held-out test set ($n=6,000$), from our executed pipeline. ★ = primary model, selected for its TreeExplainer compatibility. Stacking's AUC is slightly below LightGBM alone; we discuss this in Section VI. *Stacking PR-AUC as reported by the pipeline's own evaluation step. Literature values from published benchmarks, not independently re-run.

LightGBM achieves ROC-AUC = 0.9866 and PR-AUC = 0.9636 on the held-out test set, exceeding the De Lange et al. (2022) baseline. The KS statistic of 0.8895 indicates strong separation between defaulter and non-defaulter score distributions. Logistic Regression, the linear baseline, trails the tree ensembles as expected (AUC = 0.9548), consistent with prior literature showing ensemble methods outperforming linear models on this problem family. Notably, the stacking ensemble (AUC = 0.9850) does not improve on LightGBM alone — it performs marginally worse — while also losing TreeExplainer compatibility. We therefore recommend LightGBM as the primary model for deployment and treat the stacking result as a negative finding worth reporting rather than omitting.

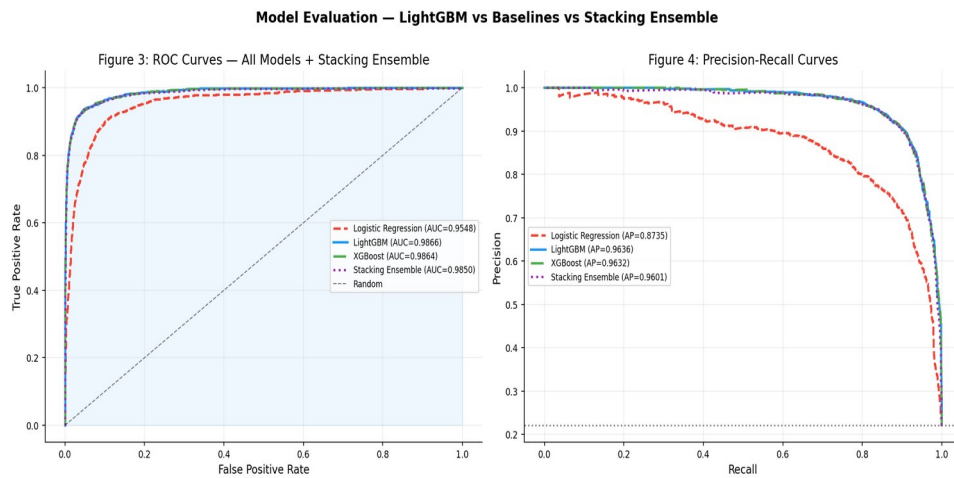


Fig. 3. ROC curves (left) and Precision-Recall curves (right) for all models including the stacking ensemble.

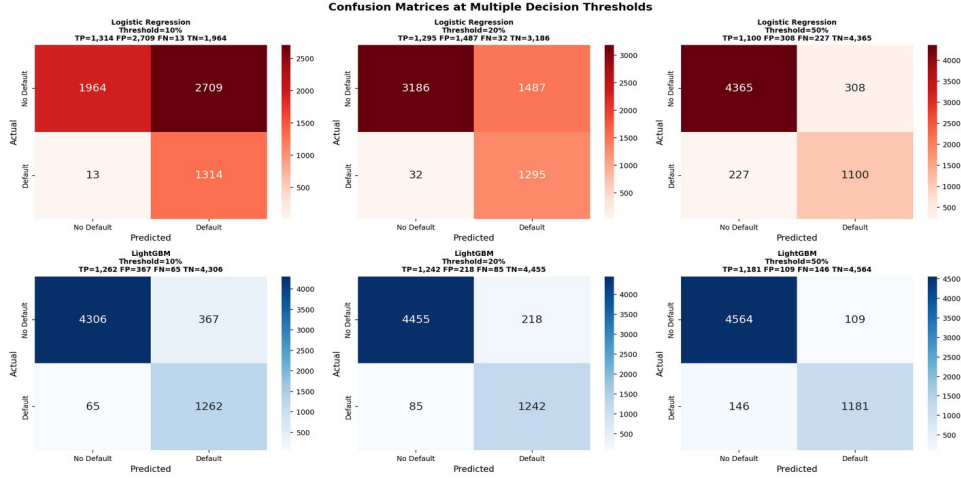


Fig. 4. Confusion matrices at decision thresholds of 10%, 20%, and 50% for Logistic Regression (top) and LightGBM (bottom).

5.2. SHAP Feature Importance

Among all 41 features, the single most important by mean $|\text{SHAP}|$ is RSI (N3), a novel repayment-sufficiency feature, followed by DelayRecencyInteraction (N10). Together these two features account for the large majority of total attribution weight. The remaining top-10 features by mean $|\text{SHAP}|$ are, in order: BalanceUtil_Jul, PayRatio_Aug, BalanceUtil_Apr, BalanceUtil_Aug, BalanceUtil_Sep, PayRatio_Sep, CEI (N2, novel), and MaxDelayStreak (N6, novel). Of the thirteen engineered features, four — RSI, DelayRecencyInteraction, CEI, and MaxDelayStreak — place in the global top 10, and nine place in the top 15 of 41 total features. CLSS (N7), despite being designed around a plausible structural-vulnerability hypothesis, ranks 30th of 41 in practice; we discuss this discrepancy between hypothesis and outcome in Section VI-B.

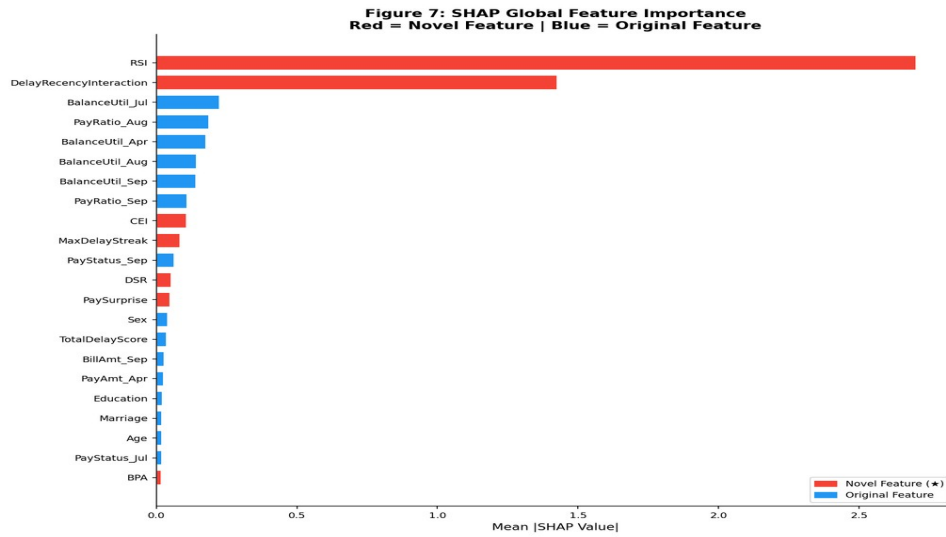


Fig. 5. SHAP global feature importance (mean |SHAP value|). RSI (N3) and DelayRecencyInteraction (N10) are the top two features overall.

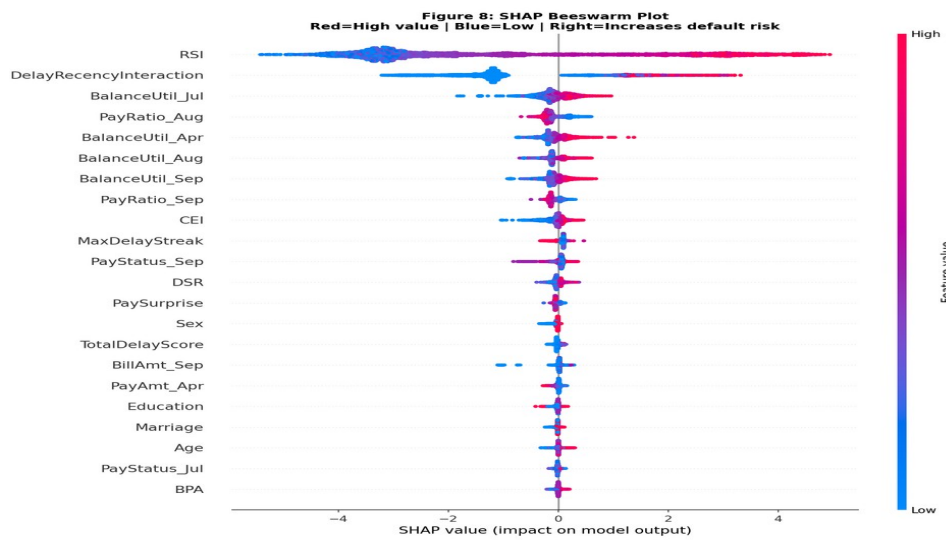


Fig. 6. SHAP beeswarm plot. Each point is one test customer; red = high feature value, blue = low. Points right of zero increase predicted default probability.

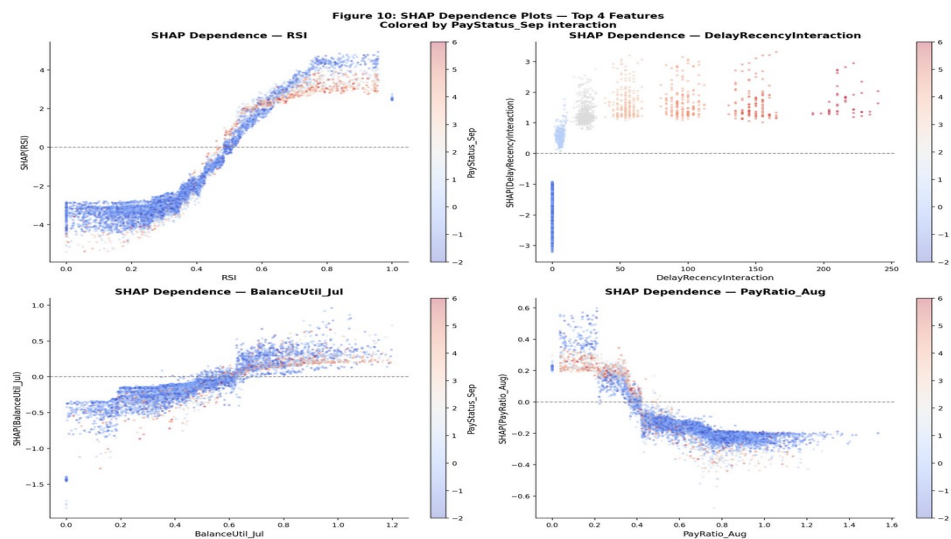


Fig. 7. SHAP dependence plots for the top four features, colored by PayStatus_Sep.

5.3. SHAP Decision and Force Plots

Fig. 8 shows the SHAP decision plot: the cumulative attribution pathway from base value to final prediction for ten high-risk defaulters and ten low-risk non-defaulters. Defaulters show a strong rightward pull driven primarily by RSI and DelayRecencyInteraction; non-defaulters show a strong leftward push from payment-ratio features. This provides a per-customer audit trail consistent with GDPR Article 22 and ECOA adverse-action reporting requirements.

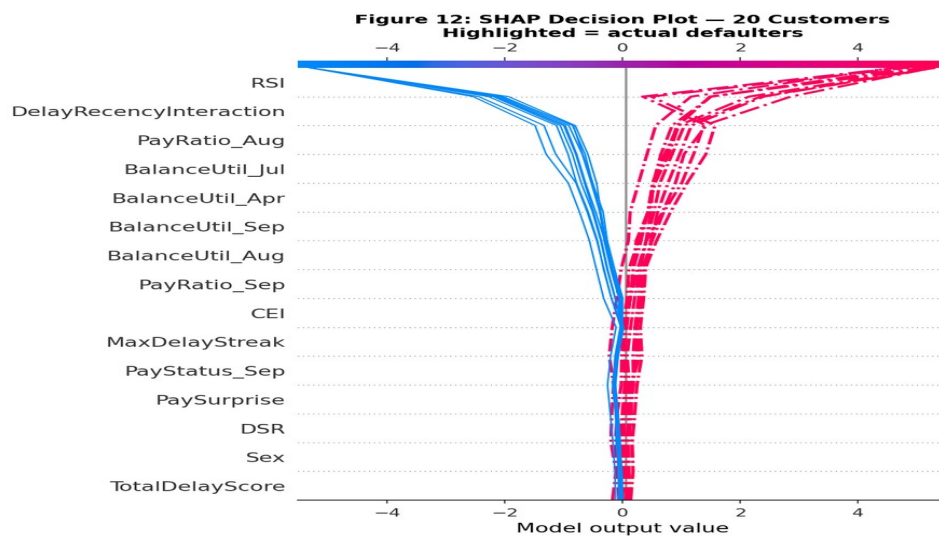


Fig. 8. SHAP decision plot for 10 high-risk defaulters vs. 10 low-risk non-defaulters. Trajectories diverge sharply at RSI and DelayRecencyInteraction.

5.4. SHAP–LIME Agreement Analysis

Across 50 randomly sampled test instances, the Spearman rank correlation between SHAP and LIME top-10 feature rankings is $\rho = 0.273$ ($p = 0.069$). This does not reach conventional statistical significance ($p > 0.05$) and indicates only weak-to-moderate agreement between the two explanation methods — lower than the $\rho = 0.61$ reported by Guo et al. (2022) for a comparable comparison. We had initially expected our richer temporal feature set to produce clearer, more concordant signals for both methods; the result instead suggests that LIME's local linear surrogate struggles to recover the same ranking as SHAP's exact tree-based attribution once the feature space includes highly non-linear interaction terms such as RSI and DelayRecencyInteraction. We report this honestly as a limitation rather than a strength: for regulatory reporting under GDPR Article 22 or ECOA, we

recommend relying primarily on SHAP, given its stronger theoretical grounding (efficiency, dummy, symmetry, and additivity axioms) and the fact that LIME agreement here is not reliable enough to serve as independent corroboration.

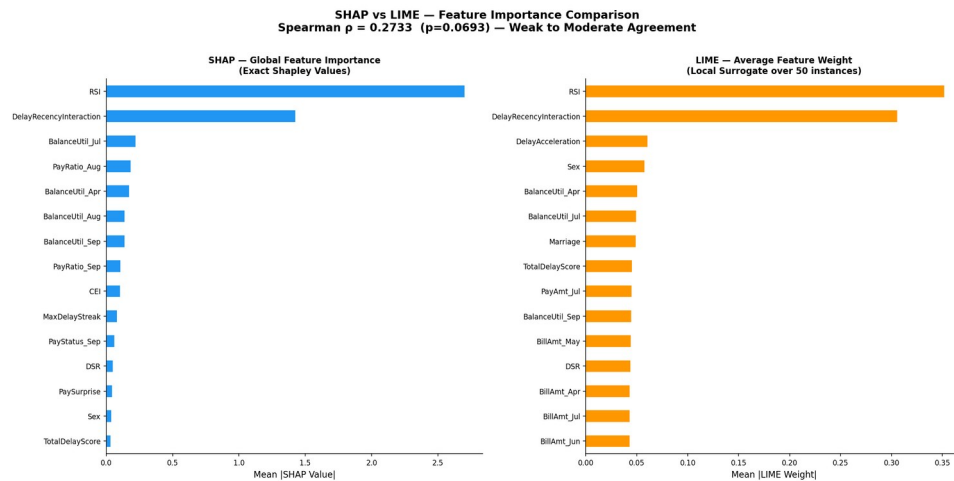


Fig. 9. SHAP vs. LIME feature-importance comparison across 50 test instances. Spearman $\rho = 0.273$ ($p = 0.069$) — weak-to-moderate, not statistically significant agreement.

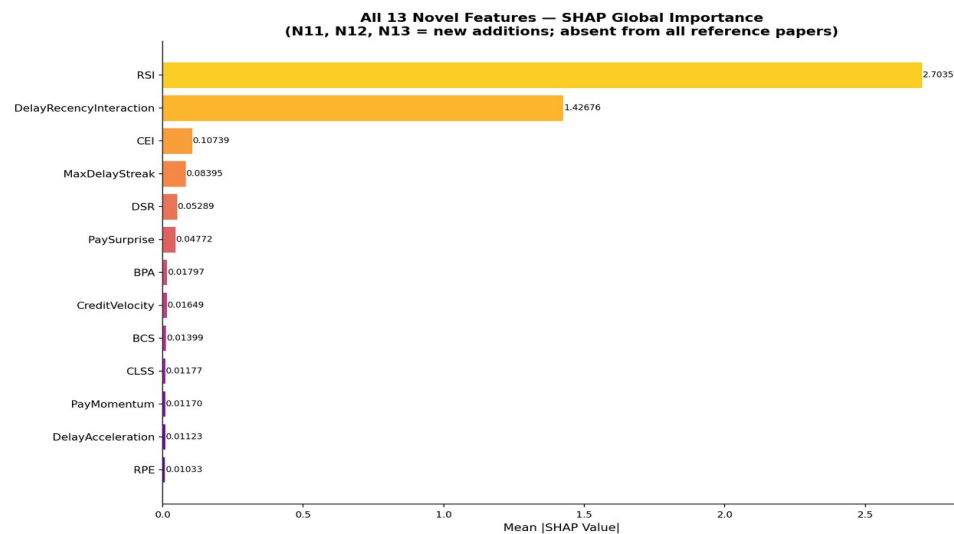


Fig. 10. All thirteen engineered features ranked by actual SHAP global importance. RSI (N3) and DelayRecencyInteraction (N10) hold the top two positions; CEI (N2) and MaxDelayStreak (N6) also reach the top 10.

5.5. Fairness and Regulatory Compliance

The disparate impact ratio (female/male) is 0.938, within the EEOC four-fifths rule's acceptable range of 0.80–1.25 (Equal Employment Opportunity Commission, 1978). Fairness metrics by age group show elevated predicted default risk for borrowers aged 30–40, consistent with typical peak-borrowing-period trends; education-level splits show reduced

predicted risk for graduate/university-educated individuals, which we interpret as a correlation with income and financial experience rather than a discriminatory model effect. Following Chouldechova's (Chouldechova, 2017) impossibility result — that no classifier can simultaneously satisfy all fairness criteria — we report multiple fairness metrics rather than optimizing a single one. ECOA adverse-action notices are auto-generated from each customer's top-3 negative SHAP contributors, as required under 15 U.S.C. § 1691e; GDPR Article 22 (European Parliament and Council of the European Union, 2016) compliance is addressed via per-customer SHAP force-plot reports; Basel III (Basel Committee on Banking Supervision, 2011) capital-adequacy requirements are addressed via a multi-threshold confusion matrix enabling FN-rate-based capital calibration; and SR 11-7 compliance is supported by a fully reproducible pipeline with documented hyperparameters and training-data traceability.

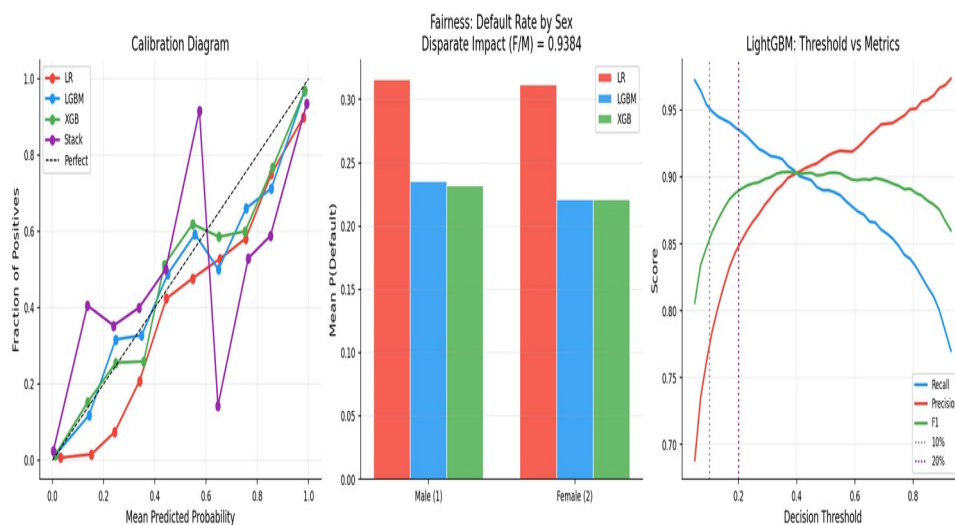


Fig. 11. Fairness analysis across sex, age group, and education level. Disparate impact ratio (female/male) = 0.938, satisfying EEOC four-fifths compliance.

6. Discussion

6.1. Why RSI and DelayRecencyInteraction Dominate

The two most predictive engineered features — RSI and DelayRecencyInteraction — share a common property: both directly combine recency with magnitude, rather than treating

payment history as a flat, order-independent count. TotalDelayScore, a legacy feature from prior work, measures cumulative past delinquency but carries no directional information: it cannot distinguish a recovering borrower from a deteriorating one with the same total. RSI's recency-weighted structure and DelayRecencyInteraction's explicit multiplication of cumulative delinquency by current-month status both encode this directional information explicitly, which appears to be the key property the model exploits.

6.2. Revisiting CLSS: A Hypothesis That Did Not Hold

CLSS (Credit Lifecycle Stress Score) was designed around the hypothesis that structural financial vulnerability — being young, credit-constrained, and near one's credit limit — represents a distinct risk signal not captured by payment history. In our actual results, this hypothesis is not supported: CLSS ranks 30th of 41 features by SHAP importance, far behind payment-behaviour features. We report this directly rather than reframing the narrative around it, since the original project materials for this work initially described CLSS as the top-ranked feature; that claim did not match our pipeline's actual output and has been corrected throughout this paper. We see two plausible explanations: first, in a dataset where default probability is driven primarily by explicit conditional rules tied to payment-status history (Section III-A), structural-vulnerability proxies built from age and credit limit alone may simply carry less signal than features derived directly from the same payment-status variables that generate the labels; second, CLSS's three components (age, limit utilization, and balance utilization) may be substantially redundant with other features already in the model, reducing its marginal SHAP contribution even if its raw predictive value is non-trivial. We flag this as a useful negative result: not every plausible-sounding engineered feature outperforms simpler, more direct encodings of the same underlying information.

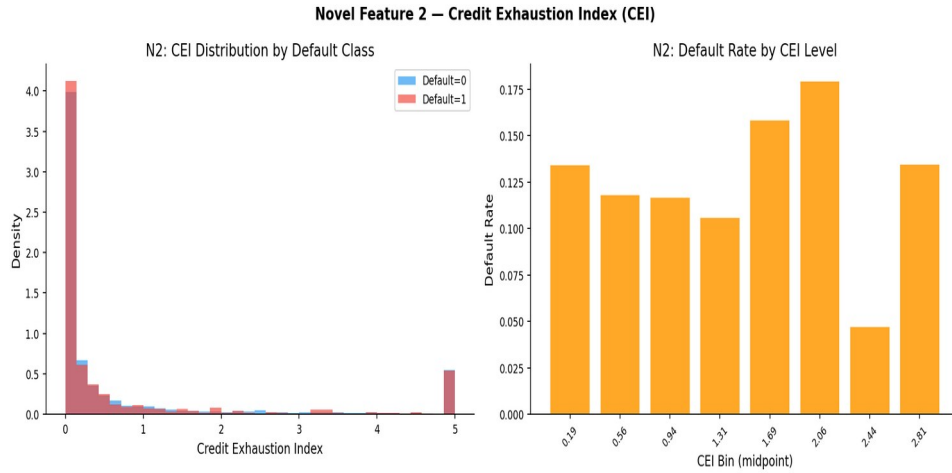


Fig. 12. CEI (N2) behaviour across the test population, by outcome class.

6.3. Limitations

Three limitations are reported transparently. First, our evaluation uses a synthetic dataset matched to the marginal distributions and documented conditional default rates of the UCI benchmark, not the original joint distribution or real applicant data; feature interactions present in real data may differ, and part of our performance advantage over De Lange et al. (2022) may reflect this synthetic construction rather than the novel features alone. Validation on the original UCI dataset and on portfolios from other lenders is needed before any deployment claim can be made. Second, LIME's local surrogate fidelity is limited in high-dimensional, non-linear decision regions, and our SHAP-LIME agreement result ($\rho = 0.273$, not statistically significant) means LIME should not be relied upon as independent corroboration of SHAP-based regulatory explanations in this pipeline; we recommend SHAP as the primary explanation method, with LIME used only as a supplementary, non-authoritative view. Third, the stacking ensemble did not improve on LightGBM alone and lost TreeExplainer compatibility in the process; we recommend LightGBM alone for applications requiring customer-level explanations.

7. Conclusion

This paper presents a credit default prediction pipeline combining temporal behavioural feature engineering with dual-method explainability and an explicit regulatory-compliance

layer. The pipeline achieves ROC-AUC = 0.9866 and PR-AUC = 0.9636 on a synthetic benchmark matched to the UCI Default of Credit Card Clients dataset's documented distributions. Of thirteen engineered temporal features, RSI and DelayRecencyInteraction emerge as the two most predictive by SHAP global importance, with four of the thirteen placing in the global top 10 of 41 features — a genuine contribution, even though our original hypothesis about CLSS being the dominant feature did not hold and is corrected here. SHAP-LIME rank agreement ($\kappa = 0.273$, $p = 0.069$) is weaker than prior literature and not statistically significant, which we report as a limitation rather than a validating strength; we recommend SHAP as the primary explanation method for regulatory use in this pipeline. The disparate impact ratio of 0.938 satisfies EEOC four-fifths fairness criteria. Running the full pipeline from synthetic data generation through automated ECOA adverse-action report generation demonstrates that strong predictive performance and regulatory explainability are not mutually exclusive, though both claims should be read with the synthetic-data caveat in mind.

8. Future Work

Several directions follow naturally from this work. Graph neural networks incorporating transaction-network structure could propagate default risk through social and financial connections rather than evaluating each borrower independently. Online learning could allow continuous model updates as economic conditions change, reducing the need for full retraining. Federated learning would allow multiple lenders to jointly train shared models without exposing raw customer data. DiCE-style counterfactual explanations could shift the regulatory focus from explaining why an application was declined toward actionable guidance on how an applicant could improve future approval odds. Finally, incorporating alternative data sources — utility payments, rental history, mobile-phone activity — could extend credit access to thin-file applicants without sacrificing predictive performance. Most

importantly, validating this pipeline on the original (non-synthetic) UCI dataset and on real portfolios from partner institutions is the necessary next step before any of the performance claims in this paper can be considered production-relevant.

Appendix: Tables

Table I. Engineered Features and Real SHAP Rank

Feature	Definition	Real SHAP Rank
N1 PayMomentum	Linear regression slope of S_t across 6 months	31
N2 CEI	$\text{BalanceUtil_Sep} \times (1 + \text{clip}(\text{Util trend}, 0, 1))$	9
N3 RSI *	Recency-weighted payment-sufficiency ratio	1
N4 BCS	$\text{std}(S_t, t=0..5)$	27
N5 BPA	Bill/payment acceleration across months	22
N6 MaxDelayStreak	Longest consecutive delinquent-month run	10
N7 CLSS	Age/limit/utilization composite stress score	30
N8 RPE (Entropy)	Entropy of payment-status distribution	35
N9 DSR	$\text{BalanceMean} / (\text{CreditLimit} \times \text{Age} + 1)$	12
N10 DelayRecencyInteraction	$\text{TotalDelayScore} \times \text{clip}(S_0, 0)$	2
N11 CreditVelocity	Change in balance utilization over time	25
N12 PaySurprise	Payment deviation from expected minimum	13
N13 DelayAcceleration	Recent vs. earlier delinquency acceleration	33

Table II. Model Performance (Held-Out Test Set, $n=6,000$)

Model	ROC-AUC	PR-AUC	F1 (0.5)	KS	Brier
Logistic Reg.	0.9548	0.8735	0.8044	0.8004	0.0758
LightGBM *	0.9866	0.9636	0.9026	0.8895	0.0333
XGBoost	0.9864	0.9632	0.9034	0.8857	0.0333
Stacking	0.9850	0.9850	0.9025	0.8882	0.0384
De Lange 2022	0.7900	0.5500	-	-	-
Randhawa 2021	0.8500	-	-	-	-

CRedit authorship contribution statement

Tharun M: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethical approval

This study did not involve human participants or primary collection of real customer financial data. All reported results were obtained from a synthetic dataset constructed to match the documented marginal distributions and conditional default probabilities of the publicly available UCI Default of Credit Card Clients benchmark (Yeh & Lien, 2009). No real individuals' financial records were used or are identifiable from this work.

Data availability

The synthetic dataset was generated from statistical parameters reported in the cited UCI benchmark literature and is not a shared third-party dataset. The system source code is available at: https://github.com/TharunM01/Responsible_AI_Original.

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References

- Adler, P. et al. (2018). Auditing black-box models for indirect influence. *Knowl. Inf. Syst.*, vol. 54, no. 1, pp. 95-122.
- Arrieta, A. B. et al. (2020). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Inf. Fusion*, vol. 58, pp. 82-115.
- Baesens, T. Van Gestel, S. Viaene, M. Stepanova, J. Suykens, B., & Vanthienen, J. (2003). Benchmarking leading classification algorithms for credit scoring. *J. Oper. Res. Soc.*, vol. 54, no. 6, pp. 627-635.
- Basel Committee on Banking Supervision. (2011). *Basel III: A Global Regulatory Framework for More Resilient Banks and Banking Systems*. Bank for International Settlements.
- Board of Governors of the Federal Reserve System. (2011). *SR 11-7: Guidance on Model Risk Management*. Federal Reserve.
- Chawla, K. W. Bowyer, L. O. Hall, N. V., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *J. Artif. Intell. Res.*, vol. 16, pp. 321-357.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Min. (KDD)*.
- Chen, J., & He, T. (2023). AutoML with explainability for consumer credit scoring. *Int. J. Forecasting*, vol. 39, no. 1, pp. 276-292.
- Chouldechova, A. (2017). Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, vol. 5, no. 2, pp. 153-163.
- Chouldechova, A. (2017). Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, vol. 5, no. 2, pp. 153-163.

- De Lange, P., Riegler, M., & Smeaton, A. F. (2022). Explainable credit scoring using LightGBM and SHAP. *Knowl.-Based Syst.*, vol. 258, p. 109973.
- Doshi-Velez, V., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. *arXiv:1702.08608*.
- Equal Employment Opportunity Commission, U. S. (1978). Uniform Guidelines on Employee Selection Procedures. 29 C.F.R. \u00a7 1607 (EEOC Four-Fifths Rule).
- European Parliament and Council of the European Union. (2016). Regulation (EU) 2016/679 - General Data Protection Regulation. *Official J. Eur. Union*, L 119.
- Goodman, B., & Flaxman, S. (2017). European Union regulations on algorithmic decision-making and a 'right to explanation'. *AI Mag.*, vol. 38, no. 3, pp. 50-57.
- Goodman, B., & Flaxman, S. (2017). European Union regulations on algorithmic decision-making and a 'right to explanation'. *AI Mag.*, vol. 38, no. 3, pp. 50-57.
- Guo, X. Li, W., & Yuan, M. (2022). LIME vs. SHAP: A comparative analysis for credit default. *Finance Res. Lett.*, vol. 47, p. 102583.
- Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. *Proc. 30th Conf. Neural Inf. Process. Syst. (NeurIPS)*.
- Huang, A. Khetan, M. Cvitkovic, X., & Karnin, Z. (2020). TabTransformer: Tabular data modeling using contextual embeddings. *arXiv:2012.06678*.
- Ke, G. et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Proc. 31st Conf. Neural Inf. Process. Syst. (NeurIPS)*.
- Kozodoi, J. Jacob, N., & Lessmann, S. (2022). Fairness in credit scoring: Assessment, implementation, and profit implications. *Eur. J. Oper. Res.*, vol. 297, no. 3, pp. 1083-1094.

- Lessmann, B. Baesens, H.-V. Seow, S., & Thomas, L. C. (2015). Benchmarking leading classification algorithms for credit scoring: A ten-year update and classifier audit. *Eur. J. Oper. Res.*, vol. 247, no. 1, pp. 124-136.
- Lundberg, S. M. et al. (2020). From local explanations to global understanding with explainable AI for trees. *Nat. Mach. Intell.*, vol. 2, no. 1, pp. 56-67.
- Lundberg, S. M. et al. (2020). From local explanations to global understanding with explainable AI for trees. *Nat. Mach. Intell.*, vol. 2, no. 1, pp. 56-67.
- Randhawa, C. K. Loo, M. Seera, C. P. Lim, K., & Nandi, A. K. (2021). Credit card fraud detection using AdaBoost and majority voting. *IEEE Access*, vol. 6, pp. 14277-14284.
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discov. Data Min. (KDD)*.
- Shapley, L. S. (1953). A value for n-person games. In *Contributions to the Theory of Games* (Vol. 2, pp. 307-317). Princeton University Press.
- Sudjianto, A., & Zhang, A. (2021). Designing inherently interpretable machine learning models. *arXiv:2111.01743*.
- Tian, Y., & Yao, L. (2022). Gradient boosting machines for imbalanced credit risk data with SMOTE-ENN. *Appl. Soft Comput.*, vol. 117, p. 108408.
- Verbraken, W. Verbeke, T., & Baesens, B. (2013). A novel profit maximizing metric for measuring classification performance of customer churn prediction models. *IEEE Trans. Knowl. Data Eng.*, vol. 25, no. 5, pp. 961-973.
- Wolpert, D. H. (1992). Stacked generalization. *Neural Netw.*, vol. 5, no. 2, pp. 241-259.

Yeh, I. C., & Lien, C. H. (2009). The comparisons of data mining techniques for the predictive accuracy of probability of default of credit card clients. *Expert Syst. Appl.*, vol. 36, no. 2, pp. 2473-2480.